Part-number correlation — approach tiers & caveats (your call)

Pick any mix. All offline, sidecar/cached, confidence-scored, review-able. Index never written.

T1 · RPSTL row parser

- Parse the parts-list table into structured rows:
- item# · SMR · NSN · CAGEC · PART# · nomenclature · qty.
- Multi-signal columns (NSN regex, 5-char CAGEC, SMR
- codes) + confidence. Host batch -> sidecar.

The core win. Recommended.

T2 · FLIS cross-validate

- PART# + CAGEC -> NSN -> INC official name (PUB LOG).
- Repairs OCR'd nomenclature; fills gaps; flags conflicts.
- Uses data already on disk. Disambiguates same part#
- across makers via CAGEC.

Accuracy lever. Recommended.

T3 · Callout localization

- OCR the figure to find the item-number balloon,
- crop tight around that callout (not the whole figure).
- Confidence-gated; falls back to the whole figure.
- Harder on dense exploded views — opt-in.

Tighter visual. Optional.

T4 · Part-number-first UX

- Search by PART NUMBER -> the exact item, its figure/callout,
- FLIS nomenclature, and Look-Alike disambiguation.
- Wildcard suffixes (e.g. -010x) and PN+CAGEC keys.
- Feeds the 3D preview + dossier + procedure views.

Makes it usable everywhere.

Hard caveats (surfaced up front)

- OCR table rows DRIFT -> mis-pairing. Fix: multi-signal
- columns + confidence + a review/override queue.
- Same part# differs by CAGEC (maker) -> key on PN+CAGEC.
- Suffix variants (-010, -010x) -> wildcard + variant grouping.
- Callout match can be wrong on busy figures -> gated + fallback.
- Big corpus -> host-side batch (can't run in the sandbox).

Recommended path

T1 (RPSTL row parser) + T2 (FLIS cross-validate) deliver the tight, accurate part#→name→figure correlation you described, using only data we already have — that's the safe, high-value core. T4 makes it searchable everywhere. T3 (callout localization) is the 'tightest visual' but the noisiest; I'd build it confidence-gated with a whole-figure fallback, or defer it. Everything ships with the usual changelog + diagram + a review queue like the sides/uncertain pattern.

Proposal diagram. Dark (R3). Tiers for part-number↔figure correlation. No build yet — awaiting your final call.